

(17.1) 식 (17.2)의 $E[L^2(K)] \leq E[L^2]$ 을 만족시키는 $K > 0$ 의 값이 항상 존재함을

증명하라.

$$\text{Var}(b_i) = \frac{\sigma^2}{1 - r_{ii}^2}, \quad i=1, 2, \dots \quad (17.2)$$

변수들은 모두 표준화된 변수이다. 즉:

$$\sum y_i = \sum x_{1i} = \sum x_{2i} = 0$$

$$\sum y_i^2 = \sum x_{1i}^2 = \sum x_{2i}^2 = 1$$

이런 경우에는 절편항 β_0 가 필요 없음을 유의하여 주절 바란다.

$$X'X = \begin{bmatrix} \sum x_{1i}^2 & \sum x_{1i}x_{2i} \\ \sum x_{2i}x_{1i} & \sum x_{2i}^2 \end{bmatrix} = \begin{bmatrix} 1 & r_{12} \\ r_{21} & 1 \end{bmatrix}$$

$$X'y = \begin{bmatrix} \sum x_{1i}y_i \\ \sum x_{2i}y_i \end{bmatrix} = \begin{bmatrix} r_{1y} \\ r_{2y} \end{bmatrix}$$

r_{12} : 변수 x_1 과 x_2 사이의 표준상관계수

r_{iy} : 변수 x_i 와 y 간의 표준상관계수

$$E[L^2] = E[(b - \beta)'(b - \beta)]$$

$$b = (X'X)^{-1}X'y \quad (\text{OLS estimator})$$

$$b(K) = (X'X + KI)^{-1}X'y, \quad K > 0 \quad (\text{Ridge estimator})$$

} start here

$$\begin{aligned} E[L^2(K)] &= E[(b(K) - \beta)'(b(K) - \beta)] = \text{var}(\hat{\beta}(K)) + \text{bias}(\hat{\beta}(K))^2 \\ &= \sigma^2 \text{tr}[(X'X + KI)^{-1}X'X(X'X + KI)^{-1}] + K^2\beta'(X'X + KI)^{-2}\beta \\ &= \sigma^2 \sum_{i=1}^p \lambda_i(\lambda_i + K)^{-2} + K^2\beta'(X'X + KI)^{-2}\beta \end{aligned}$$

$$\begin{aligned} E[L^2] &= E[(b - \beta)'(b - \beta)] = E[\text{trace}((b - \beta)'(b - \beta))] \\ &= E[\text{trace}((b - \beta)(b - \beta)')] \\ &= \text{trace}(E[(b - \beta)(b - \beta)']) \\ &= \text{trace}(\text{Var}(b)) \\ &= \sum_{i=1}^p \text{Var}(b_i) \\ &= \sigma^2 \sum_{i=1}^p \lambda_i^{-1} \end{aligned}$$

$$* b = (X'X)^{-1}X'y$$

$$\text{Var}(b) = \sigma^2(X'X)^{-1} = \sigma^2 P \Lambda^{-1} P^T, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$$

$$\begin{aligned} \text{trace}(\text{Var}(b)) &= \sigma^2 \text{trace}(\Lambda^{-1} P^T P) \\ &= \sigma^2 \text{trace}(\Lambda^{-1}) = \sigma^2 \sum_{i=1}^p \lambda_i^{-1} \end{aligned}$$

$$\begin{aligned} E[L^2(K)] &= \sigma^2 \sum_{i=1}^p \lambda_i(\lambda_i + K)^{-2} + K^2\beta'(X'X + KI)^{-2}\beta \\ &= \sigma^2 \sum_{i=1}^p \lambda_i(\lambda_i + K)^{-2} + K^2\beta'(P \Lambda P^T + K P P^T)^{-2}\beta \\ &= \sigma^2 \sum_{i=1}^p \lambda_i(\lambda_i + K)^{-2} + K^2\beta'P(\Lambda + KI)^{-2}P^T\beta \\ &= \sigma^2 \sum_{i=1}^p \frac{\lambda_i}{(\lambda_i + K)^2} + K^2 \sum_{i=1}^p \frac{1}{(\lambda_i + K)^2} a_i^2, \quad P^T\beta = P^T[\beta_1, \dots, \beta_p]^T = [a_1, \dots, a_p]^T \end{aligned}$$

$$\begin{aligned} \frac{\partial}{\partial K} E[L^2(K)] &= \frac{\partial}{\partial K} \sigma^2 \sum_{i=1}^p \frac{\lambda_i}{(\lambda_i + K)^2} + K^2 \sum_{i=1}^p \frac{1}{(\lambda_i + K)^2} a_i^2 \\ &= -2\sigma^2 \sum_{i=1}^p \frac{\lambda_i}{(\lambda_i + K)^3} + 2K \sum_{i=1}^p \frac{1}{(\lambda_i + K)^2} a_i^2 - 2K^2 \sum_{i=1}^p \frac{1}{(\lambda_i + K)^3} a_i^2 \\ \therefore \frac{\partial}{\partial K} E[L^2(K)] \Big|_{K=0} &= -2\sigma^2 \sum_{i=1}^p \frac{1}{\lambda_i^2} < 0 \end{aligned}$$

$\sigma^2 > 0$ 이고 $\lambda_i > 0$ 이므로, $K=0$ 에서 MSE 의 기울기는 "엄격한 음수 (Strictly negative)" 입니다.

함수가 $K=0$ 에선 연속이고 미분가능하며 기울기가 음수라는 것은, 아주 작은 양수 K 을 증가시키는 초기구간에서는 편향(bias)이 증가하는 속도보다 분산이 감소하는 속도가 압도적으로 빠르다는 것을 의미합니다.

따라서 평균값의 정리의 (Mean Value Theorem) 또는 테일러 전개 (Taylor expansion)에 의해, $E[L^2(K)] < E[L^2(0)] = E[L^2]$ 을 만족하는 엄격한 양수 $K > 0$ 가 항상 존재합니다.

17.2) 능형회귀(Ridge Regression)의 총평균제곱오차 $E[L^2(K)]$ 가 식 (17-17)과 같이 됨을 상세히 밝히고 그 의미

$$\frac{dE[L^2(K)]}{dK} = -2\sigma^2 \sum_{i=1}^p \frac{\lambda_i}{(\lambda_i + K)^3} + 2K \sum_{i=1}^p \frac{\lambda_i a_i^2}{(\lambda_i + K)^3}$$

이 됨을 증명하라.

$$\begin{aligned} * y &= X\beta + \epsilon = XP\bar{P}\beta + \epsilon = (XP)(P\bar{P}\beta) + \epsilon \\ &= Z\alpha + \epsilon \end{aligned}$$

$$\begin{aligned} Z &= XP, \alpha = P\bar{P}\beta, \hat{\alpha} = (Z'Z)^{-1}Z'y = (P'X'XP)^{-1}P'X'y = P'P\Lambda^{-1}P'P \cdot P'X'y \\ \hat{\alpha}(K) &= (Z'Z + KI_n)^{-1}Z'y = (P'X'XP + KI_n)^{-1}P'X'y \\ &= P'P(\Lambda + KI_n)^{-1}P'P P'X'y \\ &= (\Lambda + KI_n)^{-1}P'X'y = (\Lambda + KI_n)^{-1}Z'y \end{aligned}$$

$$\begin{aligned} Z'Z &= (XP)'XP = P'X'XP = P'P\Lambda P'P \\ &= \Lambda \end{aligned}$$

$$\begin{aligned} E[L^2(K)] &= E[(\hat{\alpha}(K) - \alpha)'(\hat{\alpha}(K) - \alpha)] \\ &= \text{tr}\{\text{Var}[\hat{\alpha}(K)]\} + \text{bias}(\hat{\alpha})^2 \\ &= \sigma^2 \text{tr}[(\Lambda + KI)^{-1}]^2 + \hat{\alpha}'[(\Lambda + KI)^{-1}\Lambda - I]'\{(\Lambda + KI)^{-1}\Lambda - I\}\hat{\alpha} \\ &= \sigma^2 \sum_{i=1}^p \frac{\lambda_i}{(\lambda_i + K)^2} + \sum_{i=1}^p \frac{K^2 a_i^2}{(\lambda_i + K)^2} \quad \dots (17-17) \end{aligned}$$

$$\Rightarrow E[L^2(K)] = E[(b(K) - \beta)'(b(K) - \beta)]$$

$$\begin{aligned} b(K) &= (X'X + KI_p)^{-1}X'y \\ &= (P\Lambda P' + KPP')^{-1}X'y \\ &= (P(\Lambda + KI_p)P')^{-1}X'y \\ &= P(\Lambda + KI_p)^{-1}P'X'y \\ &= P(\Lambda + KI_p)^{-1}Z'y \quad (\because Z = XP) \end{aligned}$$

$$\begin{aligned} E[(b(K) - \beta)'(b(K) - \beta)] &= E[(Pb(K) - P\beta)'(Pb(K) - P\beta)] \\ &= E[(\hat{\alpha}(K) - \alpha)'(\hat{\alpha}(K) - \alpha)] \\ &= \text{tr}\{(\hat{\alpha}(K) - \alpha)'(\hat{\alpha}(K) - \alpha)\} \\ &= \text{tr}\{(\hat{\alpha}(K) - \alpha)(\hat{\alpha}(K) - \alpha)'\} \\ &= \text{tr}\{(\hat{\alpha}(K) - E[\hat{\alpha}(K)] + E[\hat{\alpha}(K)] - \alpha)(\hat{\alpha}(K) - E[\hat{\alpha}(K)] + E[\hat{\alpha}(K)] - \alpha)'\} \\ &= \text{tr}\{(\hat{\alpha}(K) - E[\hat{\alpha}(K)])(\hat{\alpha}(K) - E[\hat{\alpha}(K)])'\} \quad \dots ① \\ &+ \text{tr}\{(\hat{\alpha}(K) - E[\hat{\alpha}(K)])(E[\hat{\alpha}(K)] - \alpha)'\} \quad \dots ② \\ &+ \text{tr}\{(E[\hat{\alpha}(K)] - \alpha)(E[\hat{\alpha}(K)] - \alpha)'\} \quad \dots ③ \end{aligned}$$

① ... $\text{Var}(\hat{\alpha}(K))$

② ... 0

③ ... $\text{bias}(\hat{\alpha}(K))^2$

$$\begin{aligned} \therefore E[L^2(K)] &= \text{Var}(\hat{\alpha}(K)) + \text{bias}(\hat{\alpha}(K))^2 \\ &= \underbrace{\text{Var}[(Z'Z + KI_p)^{-1}Z'y]}_{①} + \underbrace{\{E[\hat{\alpha}(K)] - \alpha\}^2}_{②} \end{aligned}$$

$$\begin{aligned} ① : (Z'Z + KI_p)^{-1}Z'Z(Z'Z + KI_p)^{-1}\sigma^2 \\ (\Lambda + KI_p)^{-1}\Lambda(\Lambda + KI_p)^{-1}\sigma^2 \\ = \sigma^2 \sum_{i=1}^p \frac{\lambda_i}{(\lambda_i + K)^2} \end{aligned}$$

$$② \{E[\hat{\alpha}(K)] - \alpha\}^2$$

$$\begin{aligned} &= \{E[(Z'Z + KI_p)^{-1}Z'y] - \alpha\}^2 \\ &= \{(Z'Z + KI_p)^{-1}Z'X\beta - \alpha\}^2 \\ &= \{(\Lambda + KI_p)^{-1}P'X'X\beta - \alpha\}^2 \\ &= \{(\Lambda + KI_p)^{-1}P'P\Lambda P'P\beta - \alpha\}^2 \\ &= \{(\Lambda + KI_p)^{-1}\Lambda\alpha - \alpha\}^2 \\ &= \{(\Lambda + KI_p)^{-1}\Lambda - I\}^2 \alpha^2 \end{aligned}$$

$$\begin{aligned} &= \sum_{i=1}^p \left\{ \frac{\lambda_i - \lambda_i - K}{\lambda_i + K} \right\}^2 a_i^2 \\ &= \sum_{i=1}^p \frac{K^2 a_i^2}{(\lambda_i + K)^2} \end{aligned}$$

$$\therefore ① + ② = \sigma^2 \sum_{i=1}^p \frac{\lambda_i}{(\lambda_i + K)^2} + \sum_{i=1}^p \frac{K^2 a_i^2}{(\lambda_i + K)^2}$$

$$\therefore E[L^2(K)] = \sigma^2 \sum_{i=1}^p \frac{\lambda_i}{(\lambda_i + K)^2} + \sum_{i=1}^p \frac{K^2 a_i^2}{(\lambda_i + K)^2}$$

$$\begin{aligned} \frac{\partial}{\partial K} E[L^2(K)] &= \sigma^2 \sum_{i=1}^p -2 \frac{\lambda_i}{(\lambda_i + K)^3} + \sum_{i=1}^p \frac{2K a_i^2}{(\lambda_i + K)^3} + \sum_{i=1}^p -2 \frac{K^2 a_i^2}{(\lambda_i + K)^3} \\ &= -2 \sum_{i=1}^p \frac{\lambda_i \cdot \sigma^2}{(\lambda_i + K)^3} + \sum_{i=1}^p \frac{2K a_i^2 (\lambda_i + K)}{(\lambda_i + K)^3} - 2 \sum_{i=1}^p \frac{K^2 a_i^2}{(\lambda_i + K)^3} \\ &= -2 \sum_{i=1}^p \frac{\lambda_i \sigma^2}{(\lambda_i + K)^3} + \sum_{i=1}^p \frac{2K \lambda_i a_i^2}{(\lambda_i + K)^3} \quad \square \end{aligned}$$

17.3) 식 (17.19)의 일반화형회귀추정량 $b(K) = (X'X + K)^{-1}X'y$ 가 사용될 때에 증평관계증
요사 $E[L^2(K)]$ 가 $K_1 = \frac{\sigma^2}{\alpha_1^2}$ 에서 최소가 됨을 증명하라.

$$\begin{aligned} L^2(K) &= (b(K) - b)'(b(K) - b) \\ E[L^2(K)] &= E[(b(K) - b)'(b(K) - b)] \\ &= E[\text{tr}((b(K) - b)(b(K) - b)')] \\ &= \text{tr}(E[(b(K) - b)(b(K) - b)']) \\ &= \text{tr}(\text{Var}(b(K))) + \text{bias}(b(K))^2 \end{aligned}$$

$$\begin{aligned} \frac{\partial}{\partial K} E[L^2(K)] &= -2 \sum_i \frac{\lambda_i \sigma^2}{(\lambda_i + K)^3} + \frac{2}{T} \sum_i \frac{K \lambda_i d_i^2}{(\lambda_i + K)^3} \\ &= -2 \sum_i \frac{\lambda_i}{(\lambda_i + K)^3} (\sigma^2 - K d_i^2) \end{aligned}$$

$$\text{if } K = \frac{\sigma^2}{\alpha_1^2}, \quad \frac{\partial}{\partial K} E[L^2(K)] = 0$$

$$\therefore E[L^2(K)] \text{ is minimum at } K = \frac{\sigma^2}{\alpha_1^2}$$

17.4) 식 (17.27)이 설명됨을 증명하라.

$$b_g = (I - P_S P_S') b \quad \dots (17.27)$$

<추정분 회귀 (Principle Component Regression)>

$$\begin{aligned} y &= X\beta + \epsilon = Z\alpha + \epsilon \quad \dots (17.22) \\ \hat{\alpha} &= Z'y \quad \dots (17.23) \end{aligned} \quad \left. \vphantom{\begin{aligned} y &= X\beta + \epsilon = Z\alpha + \epsilon \\ \hat{\alpha} &= Z'y \end{aligned}} \right\} *$$

$$\begin{aligned} \beta \text{의 최소제곱추정량은: } b &= P\hat{\alpha} \quad \dots (17.24) \\ b &= \hat{\beta} = PP'\hat{\beta} = P\hat{\alpha} \end{aligned}$$

$$\begin{aligned} P &= (P_g, P_s) \\ \alpha' &= (\alpha_g', \alpha_s') \end{aligned}$$

$$b_g = P_g \hat{\alpha}_g = P_g \Lambda_g P_g' X'y$$

$$* \quad y = X\beta + \epsilon = XP\beta + \epsilon = Z\alpha + \epsilon \quad (\because XP = Z, P'\beta = \alpha)$$

$$\begin{aligned} * \quad \hat{\alpha} &= \Lambda P'\hat{\beta} = \Lambda P'(X'X)^{-1}X'y \\ &= \Lambda P'(P\Lambda P')^{-1}X'y \\ &= \Lambda P'P\Lambda^{-1}P'X'y \\ &= P'X'y = (XP)'y = Z'y \end{aligned}$$

(Solution)

$$P = [P_1 \ P_2] \quad PP' = [P_1 \ P_2] \begin{bmatrix} P_1' \\ P_2' \end{bmatrix} = P_1P_1' + P_2P_2' = I_n$$

$$P'P = \begin{bmatrix} P_1' \\ P_2' \end{bmatrix} [P_1 \ P_2] = \begin{bmatrix} P_1'P_1 & P_1'P_2 \\ P_2'P_1 & P_2'P_2 \end{bmatrix} = \begin{bmatrix} I_1 & 0 \\ 0 & I_2 \end{bmatrix}$$

$$\hat{\alpha} = P'b = \begin{bmatrix} P_1' \\ P_2' \end{bmatrix} b = \begin{bmatrix} P_1'b \\ P_2'b \end{bmatrix}$$

$$\begin{aligned} \hat{\alpha} = \begin{bmatrix} \hat{\alpha}_1 \\ \hat{\alpha}_2 \end{bmatrix} = \begin{bmatrix} P_1'b \\ P_2'b \end{bmatrix} \quad \therefore \hat{\alpha}_1 = P_1'b \quad (\text{alternatively } \hat{\alpha} = (Z'Z)^{-1}Z'y = (P'X_c'X_cP)^{-1}P'X_c'y \\ = P'(X_c'X_c)^{-1}X_c'y = P'b) \\ \text{we also know that } P_1\hat{\alpha}_1 = b_1 \quad \therefore b_1 = P_1\hat{\alpha}_1 = P_1P_1'b = P_1P_1'b = (I - P_1P_1')b \quad \therefore P\hat{\alpha} = b \end{aligned}$$

17.5) 식 (17.28), (17.29), (17.30)을 증명하라.

$$(a) E[b_g] = \beta - \sum_{i=g+1}^p (P_i' \beta) P_i \quad \dots (17.28)$$

$$(b) \text{Var}(b_g) = \sigma^2 \sum_{i=1}^g \lambda_i^{-1} P_i P_i' \quad \dots (17.29)$$

$$(c) E[(b_g - \beta)'(b_g - \beta)] = \sigma^2 \sum_{i=1}^g \lambda_i^{-1} + \sum_{i=g+1}^p (P_i' \beta)^2 \quad \dots (17.30)$$

(Solution)

$$(a) b_g = P_g \hat{\alpha}_g = P_g P_g' b = (I - P_S P_S') b$$

$$E[b_g] = E[(I - P_S P_S') b] = \beta - P_S P_S' \beta$$

$$P_S = [P_{g+1} \ \dots \ P_p]$$

$$\therefore P_S P_S' = P_{g+1} P_{g+1}' + \dots + P_p P_p'$$

$$\therefore E[b_g] = \beta - \sum_{i=g+1}^p P_i P_i' \beta$$

this is scalar

$$= \beta - \sum_{i=g+1}^p (P_i' \beta) P_i$$

$$(b) \text{Var}(b_g) = \text{Var}((I - P_S P_S') b)$$

$$\begin{aligned} &= (I - P_S P_S') \text{Var}(b) (I - P_S P_S')' \\ &= \sigma^2 (I - P_S P_S') (X'X)^{-1} (I - P_S P_S')' \\ &= \sigma^2 (I - P_S P_S') (P\Lambda P')^{-1} (I - P_S P_S')' \\ &= \sigma^2 (I - P_S P_S') P\Lambda^{-1} P' (I - P_S P_S')' \\ &= \sigma^2 (I - P_S P_S') P\Lambda^{-1} P' (I - P_S P_S') \\ &= \sigma^2 \{ P\Lambda^{-1} P' - P\Lambda^{-1} P' P_S P_S' - P_S \Lambda_S' P_S' P' + P_S P_S' P\Lambda^{-1} P' P_S P_S' \} \\ &= \sigma^2 \left\{ P\Lambda^{-1} P' - P_S \Lambda_S' P_S' - P_S \Lambda_S' P_S' + P_S \Lambda_S' P_S' \right\} \\ &= \sigma^2 \left\{ \frac{P}{T} \frac{P' P'}{\lambda_i} - \sum_{i=g+1}^p \frac{P_i P_i'}{\lambda_i} \right\} \\ &= \sigma^2 \sum_{i=1}^g \frac{P_i P_i'}{\lambda_i} \quad \square \end{aligned}$$

$$\begin{aligned} \text{Var}(b_g) &= \text{Var}(P_g P_g' b) \\ &= P_g P_g' \text{Var}(b) P_g P_g' \\ &= \sigma^2 P_g P_g' (X'X)^{-1} P_g P_g' \\ (\dots) &= \sigma^2 P_g \Lambda_g^{-1} P_g' \\ &= \sigma^2 \sum_{i=1}^g \lambda_i^{-1} P_i P_i' \end{aligned}$$

$$(c) E[(b_g - \beta)'(b_g - \beta)] = \sigma^2 \sum_{i=1}^g \lambda_i^{-1} + \sum_{i=g+1}^p (P_i' \beta)^2$$

$$\begin{aligned} &E[(b_g - E[b_g]) + E[b_g] - \beta]'(b_g - E[b_g] + E[b_g] - \beta)] \\ &= E[\text{tr}\{(b_g - E[b_g]) + E[b_g] - \beta\}'(b_g - E[b_g] + E[b_g] - \beta)] \end{aligned}$$

$$= \text{tr}(\text{Var}(b_g)) + \text{bias}(b_g)^2$$

$$= \text{tr} \left(\sigma^2 \sum_{i=1}^g \frac{1}{\lambda_i} P_i P_i' \right) + \underbrace{\left\{ \beta - \sum_{i=g+1}^p (P_i' \beta) P_i - \beta \right\}^2}_{P_S P_S' \beta}$$

$$= \sigma^2 \sum_{i=1}^g \frac{1}{\lambda_i} \text{tr}(P_i P_i') + (P_S P_S')' (P_S P_S' \beta)$$

$$= \sigma^2 \sum_{i=1}^g \frac{1}{\lambda_i} \text{tr}(P_i' P_i) + \beta' P_S P_S' P_S P_S' \beta$$

$$= \sigma^2 \sum_{i=1}^g \frac{1}{\lambda_i} + \sum_{i=g+1}^p (P_i' \beta)^2 \quad \square$$

17.6) 제약조건이 $C\beta = 0$ 이 아니고 $C\beta = m$ 일 때에 제한 리귀추정량 $\tilde{\beta}$ 을 구하고 식 (17.34)의 편미의 크기가 무엇인가를 보려라.

$$E[\tilde{\beta}] - \beta = (x'x)^{-1}c'[c(x'x)^{-1}c']^{-1}c'\beta \quad \dots (17.34)$$

p. 594

"이 제약조건 아래에서 β 의 최소제곱추정량은

$$\hat{\beta} = \{I - (x'x)^{-1}c'[c(x'x)^{-1}c']^{-1}c'\}b$$

가 됨은 식 (17.29)에서 증명하였다."

$$C\beta = C(x'x)^{-1}x'x\beta = 0$$

$$\text{let } x_r^T = C(x'x)^{-1}x'^T \Leftrightarrow x_r = x(x'x)^{-1}C^T$$

$$\tilde{\beta} = \underset{C\beta=0}{\operatorname{argmin}} (y - x\beta)^T(y - x\beta)$$

$$\hat{z} = \underset{x_r^T z = 0}{\operatorname{argmin}} (y - z)^T(y - z)$$

$$\therefore z \in \ker(x_r^T) = \operatorname{im}(x_r)^{\perp}$$

$$\therefore \hat{z} = (\Pi_x - \Pi_{x_r})y \quad (\because \text{closest vector property})$$

$$x_r(x_r'x_r)^{-1}x_r^T = x(x'x)^{-1}c'[c(x'x)^{-1}c']^{-1}c'(x'x)^{-1}x'y$$

$$\therefore \tilde{\beta} = \hat{\beta} - (x'x)^{-1}c'[c(x'x)^{-1}c']^{-1}c'\hat{\beta}$$

If the restriction is $C\beta = m \Leftrightarrow C\beta = C\beta' \Leftrightarrow C(\beta - \beta') = 0$
 $\Leftrightarrow C\beta^* = 0 \quad (\beta^* = \beta - \beta')$

$$\tilde{\beta}^* = \hat{\beta}^* - (x'x)^{-1}c'[c(x'x)^{-1}c']^{-1}c'\hat{\beta}^*$$

$$\tilde{\beta} - \beta' = \hat{\beta} - \beta' - (x'x)^{-1}c'[c(x'x)^{-1}c']^{-1}c'(\hat{\beta} - \beta')$$

$$\tilde{\beta} = \hat{\beta} - (x'x)^{-1}c'[c(x'x)^{-1}c']^{-1}\{C\hat{\beta} - m\} \quad \checkmark$$

$$\text{bias}(\tilde{\beta}) = E[\tilde{\beta}] - \beta = -(x'x)^{-1}c'[c(x'x)^{-1}c']^{-1}\{C\beta - m\} \quad \checkmark$$

17.7) 제약조건이 $C\beta = 0$ 이 아니고 $C\beta = m$ 일 때 제한 리귀추정량 $\tilde{\beta}$ 의 총평균제곱오차가 무엇인지 구하려라.

$$\begin{aligned} \text{MSE}[\tilde{\beta}] &= (E[\tilde{\beta}] - \beta)^T(E[\tilde{\beta}] - \beta) \\ &= \operatorname{tr}(\operatorname{Var}(\tilde{\beta})) + \operatorname{Bias}(\tilde{\beta})^2 \end{aligned}$$

$$\begin{aligned} \operatorname{Var}(\tilde{\beta}) &= \operatorname{Var}(b - (x'x)^{-1}c'[c(x'x)^{-1}c']^{-1}(cb - m)) \\ &= \operatorname{Var}(\{I - (x'x)^{-1}c'[c(x'x)^{-1}c']^{-1}c'\}b) \quad \textcircled{1} \end{aligned}$$

$$\begin{aligned} &= \{I - (x'x)^{-1}c'[c(x'x)^{-1}c']^{-1}c'\} \operatorname{Var}(b) \{I - (x'x)^{-1}c'[c(x'x)^{-1}c']^{-1}c'\}' \\ &= \sigma^2 \{I - (x'x)^{-1}c'[c(x'x)^{-1}c']^{-1}c'\} (x'x)^{-1} \{I - (x'x)^{-1}c'[c(x'x)^{-1}c']^{-1}c'\}' \end{aligned}$$

$$\operatorname{Bias}(\tilde{\beta}) = (x'x)^{-1}c'[c(x'x)^{-1}c']^{-1}(m - cb)$$

$$\therefore \operatorname{Bias}(\tilde{\beta})^T \operatorname{Bias}(\tilde{\beta}) = (m - cb)'(c(x'x)^{-1}c')^{-1}c'(x'x)^{-1}c'[c(x'x)^{-1}c']^{-1}(m - cb) \quad \textcircled{2}$$

$$\text{MSE}(\tilde{\beta}) = \operatorname{tr}(\textcircled{1}) + \textcircled{2}$$

17-1) 식 (17.37)의 행렬 $M = \sigma^2 c'(x'x)^{-1}c' - C\beta\beta'C'$ 이 양반정치행렬이 되기 위한 필요충분조건이 식 (17.38)의 $\beta'C'[c(x'x)^{-1}c']^{-1}c'\beta \leq \sigma^2$ 임을 증명하라.

$$\sigma^2 c'(x'x)^{-1}c' - C\beta\beta'C' \text{ is positive definite matrix} \quad \begin{matrix} C \in \mathbb{R}^{p \times (k+1)} \\ \beta \in \mathbb{R}^{(k+1) \times 1} \\ C\beta \in \mathbb{R}^p \end{matrix}$$

(Forward proof $\Rightarrow$)

$$\forall x \in \mathbb{R}^p, x^T M x > 0 \Rightarrow \sigma^2 x^T c'(x'x)^{-1}c'x - x^T C\beta\beta'C'x > 0$$

$$\text{Let } \Sigma = c'(x'x)^{-1}c' \text{ and } C\beta = \theta, \beta'C' = \theta'$$

Σ : if $(x'x)^{-1}$ is positive definite and C is full rank, Σ is also a symmetric positive definite matrix.

$$\begin{aligned} x^T M x > 0 &\Rightarrow \sigma^2 x^T \Sigma x - x^T \theta \cdot \theta' x > 0 \\ &\Rightarrow \sigma^2 (x^T \Sigma x) > (x^T \theta)^2 \end{aligned}$$

$$\text{let } x = \Sigma^{-1}\theta$$

$$\begin{aligned} &\Rightarrow \sigma^2 (\theta^T \Sigma^{-1} \Sigma^{-1} \theta) > (\theta^T \Sigma^{-1} \theta)^2 \\ &\Rightarrow \sigma^2 (\theta^T \Sigma^{-1} \theta) > (\theta^T \Sigma^{-1} \theta)^2 \\ &\Rightarrow \sigma^2 > (\theta^T \Sigma^{-1} \theta) = \rho'C'[c(x'x)^{-1}c']^{-1}c'\beta \end{aligned}$$

(backward proof)

$$\text{Assumption: } \theta^T \Sigma^{-1} \theta \leq \sigma^2$$

WTS: M is positive definite. Meaning: $\forall x \in \mathbb{R}^p, (x^T \theta)^2 \leq \sigma^2 x^T \Sigma x$

$$x^T \theta = x^T I \theta = x^T \Sigma^{1/2} \Sigma^{-1/2} \theta = (x^T \Sigma^{1/2}) (\Sigma^{-1/2} \theta)$$

$$\text{Let } u = \Sigma^{1/2} x \text{ and } v = \Sigma^{-1/2} \theta$$

$$\text{Then } x^T \theta = u^T v.$$

Apply the Cauchy-Schwarz Inequality: $(u^T v)^2 \leq (u^T u)(v^T v)$

$$(x^T \theta)^2 \leq (x^T \Sigma^{1/2} \Sigma^{1/2} x) (\theta^T \Sigma^{-1/2} \Sigma^{-1/2} \theta) = (x^T \Sigma x) (\theta^T \Sigma^{-1} \theta)$$

However, in our assumption, $\theta^T \Sigma^{-1} \theta \leq \sigma^2$. Thus,

$$(x^T \theta)^2 \leq \sigma^2 (x^T \Sigma x) \quad 0$$

17. 식 (17-39)의 축소추정량 $\tilde{\beta}_s = cb$ 의 총 평균제곱오차를 구하라.

p. 597.

... 편향의 추정 이외에도 Stein (17-20)이 고안한 축소추정량 (Shrunken estimator)이란 것도 있는데, 이는 최소제곱추정량 b 를 일괄적으로 축소시킨 추정량

$$\tilde{\beta}_s = cb, \quad 0 < c < 1 \quad (17-39)$$

이다.

$$\begin{aligned} \text{MSE}(\tilde{\beta}_s) &= E[(\tilde{\beta}_s - \beta)^T(\tilde{\beta}_s - \beta)] \\ &= E[(\tilde{\beta}_s - E[\tilde{\beta}_s]) + E[\tilde{\beta}_s] - \beta)^T(\tilde{\beta}_s - E[\tilde{\beta}_s]) + E[\tilde{\beta}_s] - \beta)] \\ &= E[(\tilde{\beta}_s - E[\tilde{\beta}_s])(\tilde{\beta}_s - E[\tilde{\beta}_s])^T] + E[(E[\tilde{\beta}_s] - \beta)(E[\tilde{\beta}_s] - \beta)^T]] \\ &= \text{tr}(\text{Var}(\tilde{\beta}_s)) + \text{bias}(\tilde{\beta}_s)^T \text{bias}(\tilde{\beta}_s) \\ &= \text{tr}(c^2 \text{Var}(b)) + (c-1)^2 \beta^T \beta \\ &= \sigma^2 c^2 + \text{tr}(c^2 X^T X) + (c-1)^2 \sum_{i=1}^p \beta_i^2 \\ &= \sigma^2 c^2 + \text{tr}(P \Lambda^{-1} P^T) + (c-1)^2 \sum_{i=1}^p \beta_i^2 \\ &= \sigma^2 c^2 \sum_{i=1}^p \lambda_i^{-1} + (c-1)^2 \sum_{i=1}^p \beta_i^2 \end{aligned}$$

$$(2) E[\hat{y}_k(z)] - z'a$$

$$\hat{y}_k(z) = z' \hat{a}(k), \quad E[\hat{y}_k(z)] = z' E[\hat{a}(k)]$$

$$\begin{aligned} \hat{a}(k) &= (z'z + KI)^{-1} z'y, \quad E[\hat{a}(k)] = (z'z + KI)^{-1} z'X\beta \\ &= (\Lambda + KI)^{-1} P X' X \beta \\ &= (\Lambda + KI)^{-1} P \Lambda \beta \end{aligned}$$

$$\therefore z' E[\hat{a}(k)] = z' (\Lambda + KI)^{-1} P \Lambda \beta$$

$$\begin{aligned} E[\hat{y}_k(z)] - z'a &= z' P (\Lambda + KI)^{-1} P \Lambda \beta - z' \beta \\ &= z' \{ P (\Lambda + KI)^{-1} P \Lambda - I \} \beta \quad \checkmark \end{aligned}$$

$$(3) \text{MSE}[\hat{y}_k(z)]$$

$$\begin{aligned} &= E[\{\hat{y}_k(z) - y\} \{\hat{y}_k(z) - y\}^T] \\ &= \underbrace{\text{Var}[\hat{y}_k(z)]}_{(1)} + \underbrace{\text{bias}[\hat{y}_k(z)] \text{bias}[\hat{y}_k(z)]^T}_{(2)^2} \end{aligned}$$

$$\begin{aligned} \therefore \text{MSE}[\hat{y}_k(z)] &= \sigma^2 z' P \Lambda (\Lambda + KI)^{-1} P' z' + \\ &\quad z' \{ P (\Lambda + KI)^{-1} P \Lambda - I \} \beta \beta^T \{ P (\Lambda + KI)^{-1} P \Lambda - I \}^T z' \end{aligned}$$

z, z' 주의

17.10 식 (17-13)의 선형 모형, $y = Za + \epsilon$ 에 대하여 능력추정량

$$\hat{a}(k) = (z'z + KI)^{-1} z'y$$

가 사용될 때, 어떤 임의의 점 $z' = x'P$ 에서의 반응의 추정량

$$\hat{y}_k(z) = z' \hat{a}(k)$$

를 생각하여 보자. 다음을 구하라.

$$(1) \text{Var}[\hat{y}_k(z)] \quad (2) E[\hat{y}_k(z)] - z'a \quad (3) \text{MSE}[\hat{y}_k(z)]$$

$$\begin{aligned} (1) \text{Var}[\hat{y}_k(z)] &= \text{Var}(z' \hat{a}(k)) = z' \text{Var}(\hat{a}(k)) z \\ &= z' \text{Var}((z'z + KI)^{-1} z'y) z \\ &= z' (z'z + KI)^{-1} z' \text{Var}(y) z (z'z + KI)^{-1} z' \end{aligned}$$

$$y = X\beta + \epsilon = X P P' \beta + \epsilon \quad \therefore Z = X P, \quad a = P' \beta$$

$$z'z = P' X' X P = P' P \Lambda P' P = \Lambda$$

$$\begin{aligned} \text{Var}(\hat{a}(k)) &= \text{Var}((z'z + KI)^{-1} z'y) \\ &= (z'z + KI)^{-1} z' \text{Var}(y) z (z'z + KI)^{-1} \\ &= \sigma^2 (\Lambda + KI)^{-1} \Lambda (\Lambda + KI)^{-1} \\ &= \sigma^2 \Lambda (\Lambda + KI)^{-2} \end{aligned}$$

$$\therefore z' \text{Var}(\hat{a}(k)) z = \sigma^2 z' P \Lambda (\Lambda + KI)^{-2} P' z \quad \checkmark$$

17.11) 식 (17.26)의 구성분위추정량

$$b_g = p_g \hat{a}_g$$

가 사용될 때 어떤 임의의 점 x 에서의 반응의 추정량

$$\hat{y}_g(x) = x' b_g$$

를 고려하여 보자. 다음을 구하라:

$$(1) \text{Var}[\hat{y}_g(x)] \quad (2) E[\hat{y}_g(x)] - x'\beta \quad (3) \text{MSE}[\hat{y}_g(x)]$$

$$(1) x'X = P\Lambda P^T \quad P = [p_g \ p_s], \quad \Lambda = \begin{bmatrix} \Lambda_1 & 0 \\ 0 & \Lambda_2 \end{bmatrix}$$

(Approach #1)

$$b = (X'X)^{-1}X'y = P\Lambda^{-1}P^T \cdot X'y = [p_g\Lambda_1^{-1} + p_s\Lambda_2^{-1}]Z'y = p_g\Lambda_1^{-1}Z'y + p_s\Lambda_2^{-1}Z'y \\ = p_g\Lambda_1^{-1}p_g^T x'y + p_s\Lambda_2^{-1}p_s^T x'y$$

(Approach #2)

$$y = Z_1 a_1 + Z_2 a_2 + \epsilon$$

$$\hat{a}_1 = (Z_1'Z_1)^{-1}Z_1'y = (p_g^T X'X p_g)^{-1} p_g^T X'y \\ = (p_g^T P\Lambda P^T p_g)^{-1} p_g^T X'y \\ = \Lambda_1^{-1} p_g^T X'y$$

$$\hat{a}_1 = p_g^T b_g = \Lambda_1^{-1} p_g^T X'y$$

$$\therefore b_g = p_g \Lambda_1^{-1} p_g^T X'y$$

$$\text{Var}(b_g) = p_g \Lambda_1^{-1} p_g^T X^T \sigma^2 I_n X p_g \Lambda_1^{-1} p_g^T \\ = \sigma^2 p_g \Lambda_1^{-1} p_g^T$$

$$\therefore \text{Var}(x'b_g) = x' \text{Var}(b_g) x'^T \\ = \sigma^2 x' p_g \Lambda_1^{-1} p_g^T x \\ = \sigma^2 \sum_{i=1}^g \frac{(x'p_i)^2}{\lambda_i}$$

$$(2) E[\hat{y}_g(x)] - x'\beta$$

$$= E[x'b_g] = x'E[b_g] = x'p_g\Lambda_1^{-1}p_g^T X^T \beta \\ = x'p_g\Lambda_1^{-1}p_g^T p_g \Lambda_1 p_g^T \beta \\ = x'p_g p_g^T \beta$$

$$\therefore E[\hat{y}_g(x)] - x'\beta = x'[p_g p_g^T - I] \beta$$

$$\star (3) \text{MSE}[\hat{y}_g(x)] = \text{Var}(\hat{y}_g(x)) + \text{bias}(\hat{y}_g(x))\text{bias}(\hat{y}_g(x))^T$$

$$= \sigma^2 \sum_{i=1}^g \frac{(x'p_i)^2}{\lambda_i} + (x'p_g p_g^T \beta)^2$$

$$P P^T = [p_g \ p_s] \begin{bmatrix} p_g^T \\ p_s^T \end{bmatrix} = p_g p_g^T + p_s p_s^T = I_n \\ \therefore p_g p_g^T = I_n - p_s p_s^T$$

$$\text{MSE}[\hat{y}_g(x)] = \sigma^2 \sum_{i=1}^g \frac{(x'p_i)^2}{\lambda_i} - \left[\sum_{i=g+1}^p (x'p_i)(p_i'\beta) \right]^2$$